COURIER CHARGE OPTIMIZATION

Assignment Matter • Problem Statement • Rules • Algorithm • Pseudocode

1. Problem Statement

Develop a Python program to calculate and compare courier delivery charges for a parcel. The program accepts the parcel's weight and dimensions, delivery distance zone, urgency, and restricted-item status. If the parcel is restricted, it is rejected. Otherwise, the program calculates the parcel volume, size charge, distance charge, standard charge, and express charge. Based on whether urgent delivery is requested, it selects the suitable service and displays the final delivery charge.

2. Rules and Conditions

Item	Condition / Action
Restricted item	If the user enters Y, reject the parcel and do not calculate charges.
Parcel volume	Calculate volume = length × width × height (cubic centimetres).
Size charge	Volume up to 5,000 cm³: ■20; up to 15,000 cm³: ■50; above 15,000 cm³: ■100.
Distance zone	Zone 1 (Local): ■50; Zone 2 (State): ■100; Zone 3 (National): ■200.
Standard service	Weight × ■20 + size charge + distance charge.
Express service	Weight × ■35 + size charge + distance charge.
Service selection	If urgent is Y, select Express; otherwise, select Standard.
Final output	Display both service charges, selected service, and final delivery charge.

3. Algorithm

1. Start the program.
2. Read the parcel weight, length, width, height, distance zone, urgent-delivery choice, and restricted-item choice.
3. If the parcel is restricted, display a rejection message and stop charge calculation.
4. Otherwise, calculate the parcel volume by multiplying length, width, and height.
5. Assign the size charge according to the parcel volume: ■20, ■50, or ■100.
6. Assign the distance charge according to the zone: ■50 for Local, ■100 for State, or ■200 for National.
7. Calculate the standard charge using weight × 20 + size charge + distance charge.
8. Calculate the express charge using weight × 35 + size charge + distance charge.
9. If urgent delivery is Y, select Express and set the final charge to the express charge; otherwise, select Standard.
10. Display the standard charge, express charge, selected service, and final delivery charge.
11. Stop the program.

4. Pseudocode

```
BEGIN
    INPUT weight, length, width, height
    INPUT zone, urgent, restricted

    IF restricted = "Y" THEN
        DISPLAY "Parcel rejected due to declared restrictions."
    ELSE
        volume = length * width * height

        IF volume <= 5000 THEN
            size_charge = 20
        ELSE IF volume <= 15000 THEN
            size_charge = 50
        ELSE
            size_charge = 100
        END IF

        IF zone = 1 THEN
            distance_charge = 50
        ELSE IF zone = 2 THEN
            distance_charge = 100
        ELSE
            distance_charge = 200
        END IF

        standard = (weight * 20) + size_charge + distance_charge
        express = (weight * 35) + size_charge + distance_charge

        IF urgent = "Y" THEN
            selected_service = "Express"
            final_charge = express
        ELSE
            selected_service = "Standard"
            final_charge = standard
        END IF

        DISPLAY standard, express
        DISPLAY selected_service, final_charge
    END IF
END
```

5. Concepts Used

Variables: weight, length, width, height, zone, urgent, restricted, volume, size_charge, distance_charge, standard, express, selected_service, and final_charge. Data types: float, integer, and string. Operators: multiplication (*), addition (+), comparison (<=, ==), and assignment (=). Conditional statements: if, elif, and else. String operation: upper() converts the user's Y/N response to uppercase. Formatted strings (f-strings) display the charges to two decimal places.

6. Conclusion

The Courier Charge Optimization program checks parcel restrictions, calculates size and distance charges, compares standard and express delivery costs, and selects a service based on urgency. It demonstrates the use of variables, data types, arithmetic operators, conditional statements, and string operations in Python.